Equal forces, different accelerations

LEARNING TARGET · R8 I can identify force pairs acting on different objects.

SECTION A KEY VOCABULARY

Force pair	A-on-B and B-on-A: equal and opposite.
Receivers	Two objects; use separate force diagrams.
Mass effect	Same force size: smaller mass, larger a.
System	Pair forces cancel internally for combined A+B.

SECTION B KEY EQUATIONS + UNITS

$$F_{A\to B} = -F_{B\to A}$$

equal and opposite

a_A F_A/m_A
a_B F_B/m_B

SECTION C COLOR IT · TWO RECEIVERS

*Same **6 N** force size.*

Equal force sizes on different receivers



$a_A = -6$ m/s²

$a_B = +2$ m/s²

☐ **1 · ORANGE** Force on A: left

☐ **2 · ORANGE** Force on B: right

☐ **3 · PENCIL** Objects and masses

SECTION D WORKED EXAMPLE

Cart A has mass **1 kg**; cart B has mass **3 kg**. Their interaction pushes A left and B right with **6 N** each. **Find both accelerations.**

GIVEN	EQUATION	WORK	ANSWER
m_A = **1 kg** m_B = **3 kg** right is +	$F = ma$ on each cart. No other horizontal forces.	a_A = −6/1 = **−6 m/s²** a_B = +6/3 = **+2 m/s²**	**−6; +2 m/s²** A has 3 times B's acceleration size.

SECTION E THE 2026-XJ THREAD

DAY 14 OF 22 · NEW TOOL: INTERACTION PAIRS

Earth and fictional XJ pull on each other equally at the same instant. Their very different masses give different accelerations; the pair does not cancel in XJ's own force diagram.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else